

Productivity and Broadband: The Human Factor

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Abstract

While the productivity paradox has been largely resolved, debate persists about the impacts of information and communications technologies on productivity, particularly as regards broadband Internet connections. To date, the bulk of the literature has not addressed the impacts of broadband provision on regional variations in productivity. This study develops a series of spatial econometric models that examine the link between broadband provision and productivity for US counties. The models developed not only consider the link between productivity and broadband provision, but they also evaluate the variability in broadband impacts that are related to the quality of human capital stock within counties. The results of the models are evidence of skill-based technological change and suggest that in general, broadband has a positive impact on productivity only in locales with high levels of human capital and/or highly skilled occupations.

Keywords

broadband, productivity, human capital, technological change, technology policy

Introduction

It has been over twenty years since Robert Solow declared the computer age visible everywhere but in the productivity statistics (Solow 1987). This statement referred to

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the phenomenon that despite heavy investment in telecommunications starting in the mid-1970s, the impacts of information and communications technologies (ICTs) were not evident in the statistical data until the second half of the 1990s (Jalava and Pohjola 2002; Corrado, Hulten, and Sichel 2006). As of the mid-1990s, the productivity paradox was abandoned in favor of productivity resurgence as evidence emerged in statistical data that ICTs did indeed have an enormous impact on productivity growth (Jorgenson, Ho, and Stiroh 2008).

Since the 1970s, computers and innovations in telecommunications systems have dramatically changed every aspect of life including business, education, health, and entertainment (Parham 2002). The ability of these time-shrinking technologies to facilitate the finding, sharing, and storing of information has been the source of organizational innovations, and has revolutionized business processes. A key source of this dramatic change was the increased Internet connection speeds brought about by the development of high speed, always on broadband Internet connections.

Broadband has joined prior technological advancements such as railroads, roads, and electricity as a key part of the necessary infrastructure for economic and social development (Jordan and Leon 2011). In fact, the technological dynamism and pervasive use of broadband in the last two decades warrants its classification as a general-purpose technology (GPT) along with other key technological innovations such as the steam engine and semiconductors (Bresnahan and Trajtenberg 1995). Many of the benefits accrued from broadband use stem from their enabling capability, which allow for the development of new business processes and work practices. This enabling capacity of broadband Internet connections promotes innovational complementarities in downstream sectors, which magnify the impacts of innovations in the GPT (Bresnahan and Trajtenberg 1995; Brynjolfsson and Hitt 2000). Given the magnifying impact of broadband, it is no surprise that it is the focal point of a new system for economic and social development (Jordan and Leon 2011).

The transformative nature of broadband, and its perception as a fundamental part of our national infrastructure, led to the development of the National Broadband Plan, which was released in 2010 (Federal Communications Commission [FCC] 2010). The goal of this plan was to broaden broadband access and improve its quality (especially in terms of speed) for all Americans. The plan also highlighted the role of broadband in local economic development and the need for local communities to incorporate this increasingly essential infrastructure into local economic development strategies (FCC 2010). Several studies have found that broadband does provide benefits to local economies in the form of employment gains (Gillett et al. 2006; Crandall, Lehr, and Litan 2009; Thompson and Garbacz 2008; Shideler, Badasyan, and Taylor 2007) and economic growth (Czernich et al. 2009; Koutroumpis 2009). However, the impact of broadband on productivity remains understudied. While there is a voluminous literature on the impacts of ICTs on productivity, the majority of these studies focus primarily on computer hardware (Dedrick, Gurbaxani, and Kraemer 2003) and not telecommunications infrastructure, such as that required to provide high-speed Internet access.

Given the understudied nature of the link between broadband and productivity, this study will evaluate the link between broadband and productivity for counties in the continental United States to determine whether places with the presence of broadband and/or higher levels of broadband provision experienced greater productivity gains over an eight-year study period (2000–2007). The analysis will also evaluate the role that a skilled labor force plays in the impact of broadband on regional productivity to address the complementary nature of ICTs and skilled labor (Jorgenson 2001; Parham 2002). Specifically, this study will address three primary research questions. First, does the mere presence of broadband impact productivity levels in later years? Second, is a higher level of broadband provision associated with higher regional productivity levels? Third, does the quality of human capital stock within regions magnify the impact of broadband on productivity? Of particular interest is how the initial spatial of broadband in 1999 impacted changes in productivity in later years. The initial of this now critical infrastructure may have provided regions, with the necessary human capital resources to exploit this technology, with a first-mover advantage productivity advantage. The results of the econometric models estimated in this article suggest that this appears to be true for US counties between 2000 and 2007.

Productivity Impacts of ICTs

Economic theory suggests that ICTs have two impacts on economic growth: impacts on the productivity of computer producing industries and impacts on computer-using industries (Jorgenson and Stiroh 2000; Jalava and Pohjola 2002; Parham 2002). With respect to ICT-using firms, these technologies enable the development of new business processes and new work practices. These kinds of business improvements allow firms to increase the quality of outputs and subsequent growth in multifactor productivity (Brynjolfsson and Hitt 2000). Although industry-level studies show varied impacts of ICT use across industries (Crandall, Lehr, and Litan 2009) and countries (Schreyer 2000; Colecchia and Schreyer 2001; Daveri 2002; Pohjola 2002), there is general agreement that ICTs positively impact productivity (Kelly 1994; Black and Lynch 1997; Lehr and Lichtenberg 1998; Martínez, Rodríguez, and Torres 2010).

The Productivity Paradox

The productivity paradox evident in statistical data in the late 1980s and early 1990s spawned widespread interest in uncovering the mismatch between what economists knew to be true, that technological change has positive economic benefits, and the statistical reality of slow productivity growth. Early efforts at uncovering this link proved unsuccessful, however, and no quantifiable benefits to ICTs could be found (Roach 1987; Bailey and Gordon 1988; Strausman 1990; Morrison and Berndt 1997). For example, the study by Roach (1987) examining computer investment

between 1977 and 1989 found no relationship between investment and increased output per worker (Brynjolfsson and Hitt 2000).

Later attempts at uncovering this link proved more successful and by the mid-1990s the explosive investment in ICTs that began in the mid-1970s became evident in statistical analyses (Jalava and Pohjola 2002; Corrado, Hulten, and Sichel 2006). These impacts became observable in both micro data (Brynjolfsson and Hitt 1996; Lehr and Lichtenberg 1999) and aggregate data (Oliner and Sichel 2000; Jorgenson 2001). For example, Oliner and Sichel (2000) found that the manufacturing and use of information technology have contributed to the uptick in productivity growth since 1995.

Explanations for the Productivity Paradox

A retrospective look at the productivity paradox suggests several reasons for the lack of evidence of computer impacts in the data. The lack of evidence has been attributed to lagged effects of investment (Crandall, Lehr, and Litan 2009; Pilat 2004; Martínez, Rodríguez, and Torres 2010) and measurement issues stemming from impacts in the services sector (Pilat 2004; Crandall, Lehr, and Litan 2009). Additionally, Oliner and Sichel (1994, 2000) suggested that the use of computing equipment as the measure of ICT investment represented too small a share of the total capital stock in the economy, and obtained better results by using a measure of IT investment that consisted of computer hardware and software and communications equipment. Martínez, Rodríguez, and Torres (2010) provide additional insights as to the reasons for the lagged effect of IT investment on productivity. They suggest there are organizational costs that must be absorbed by adopters of these technologies before productivity gains can be realized: after companies become accustomed to the organizational changes and new technologies, this is when the gains accrue from investment.

Another explanation for the lagged effect of IT investment is the necessary modifications to national accounts that are required to capture these effects (Brynjolfsson and Hitt 2000; Jorgenson 2001; Corrado, Hulten, and Sichel 2006; Jorgenson, Ho, and Stiroh 2008). In particular, accounting measures and price indices have been noted to lag technological change (Brynjolfsson and Hitt 2000). Precomputer growth accounting practices obscured much of the value of intangible assets such as new business processes and new production systems by treating these items as expenses rather than investments (Brynjolfsson and Hitt 2000). Prior to 1985, for example, computer prices (holding product quality constant) were not yet included in the Bureau of Economic Analysis (BEA) National Income and Product Accounts tables (Jorgenson 2001). In fact, computer prices did not find their way into the producer price index until the 1990s (Jorgenson 2001). When the government changed accounting practices to recognize software as an investment rather than an expense it had a notable impact on national statistics (Corrado, Hulten, and Sichel 2006). Brynjolfsson and Hitt (2000) found, for example, that this accounting

change added 0.15 to 0.20 points to real average gross domestic product (GDP) growth in the 1990s.

Broadband and Productivity

While the literature on ICT impacts and productivity is voluminous, little work has been done on *broadband* and its impact on productivity. The work on broadband impacts completed to date has focused largely on employment impacts of broadband (Gillet et al. 2006; Crandall, Lehr, and Litan 2009; Shideler, Badasyan, and Taylor 2007; Thompson and Garbacz 2008), intercountry comparisons of the link between broadband and productivity (Czernich et al. 2009; Koutroumpis 2009), and the impact of broadband on local exchange carriers (Majumdar 2008; Majumdar, Carare, and Chang 2010). A 2009 study by Czernich et al. looked at the impact of broadband on GDP growth between 1996 and 2007 for twenty-five Organization for Economic Co-operation and Development (OECD) countries and found a statistically significant relationship between broadband adoption and per capita GDP. A similar study by Koutroumpis (2009) leveraged a simultaneous equation approach to look at the link between broadband penetration and GDP growth between 2002 and 2007 for twenty-two OECD countries. The study found that a 1 percent increase in broadband penetration led to a 0.025 percent increase in GDP. A recent study by the European Commission (2008) also found that increased specialization in knowledge-intensive activities and broadband-based development of innovative markets resulted in a growth of the European gross value added (GVA) of €82.4 or 0.71 percent in 2006. However, this impact varied across countries depending on the level of broadband development. In the most broadband-advanced European countries (such as Scandinavia), broadband-related GVA growth reached 0.89 percent, whereas in the countries with less-developed broadband (such as the Eastern countries), this growth was limited to 0.47 percent.

Two US-based studies have utilized state-level data to evaluate the link between broadband, output and employment (Crandall, Lehr, and Litan 2009), and broadband and GDP (Thompson and Garbacz 2008). Both studies find that broadband positively impacts employment at the state level. Other US studies have looked at the impact of broadband on local exchange carriers. Majumdar (2008) uses panel data estimation techniques to examine the impact of broadband adoption on the wages and employment of US local exchange carriers between 1988 and 2001. The study finds broadband has a positive and significant impact on wages but a negative impact on local exchange carrier employment. Majumdar, Carare, and Chang (2010) use local exchange carrier data to look at the impact of broadband on the productivity of firms deploying broadband in the United States between 1995 and 2000. They find a positive relationship between broadband deployment and the productivity of the deploying firms. Perhaps, the study that most closely approximates the present study is an analysis that evaluated the impact of broadband availability on the average pay per employee for US counties between 1999 and 2006 (Kolko 2012).

An oversight of this study, however, is that it does not account for the spatial processes at work on broadband provision in the United States.

ICTs and Human Capital

Over the last two decades, the analysis of human capital has taken on a progressively more central role in discussions regarding the growth and success of nations and regions. This is primarily because advanced societies have increasingly evolved toward what has been called a “knowledge-based economy” (OECD 2006), whereby human capital is seen to be a crucial feature of economic growth (Faggian and McCann 2009a).

Since the mid-1980s, the externalities related to education and human capital have been included as key variables in models of endogenous economic growth (Romer 1986; Lucas 1988; Krugman 1991) in order to explicitly recognize the link between these externalities and higher levels of growth (Acs 2004). In fact, several studies have found positive economic impacts related to the level of human capital within regions (Rauch 1993; Glaeser, Scheinkman, and Shleifer 1995; Simon and Nardinelli 1996, 2002; Faggian and McCann 2009b). Aside from the general importance of human capital to regional economies, the evolution of the economy in recent decades from a manufacturing to a services-oriented economy has placed increased importance on the role of human capital in the economic competitiveness of cities, regions, and nations (Lloyd and Clark 2001).

Not only is human capital alone important for regional economic development, but the literature on ICTs also suggests that human capital plays a key role in the adoption and subsequent impacts of these technologies (Benhabib and Spiegel 1994). In attempting to include this link in economic theory, a variety of theoretical models that link technological skills to technological change have been developed (Stokey 1988; Chari and Hopenhayn 1991; Grossman and Helpman 1991; Young 1993). Recently, applied evaluations of the impact of IT investments note that complimentary factors are key to explaining the additional returns from IT investments (Dedrick, Gurbaxani, and Kraemer 2003). In this regard, several studies recognize that human capital is a key complimentary factor that plays a role of the productivity impacts of ICTs on regional economies (Baldwin, Diverty, and Sabourin 1995; Entorf and Kramarz 1998; Bartelsman and Doms 2000; Gretton, Gali, and Parham 2004). Entorf and Kramarz (1998), for example, found that computer technology use is higher for more skilled workers. This link between ICTs and the skill level of workers has several implications for policy. Parham (2002), for example, suggests ICT policies should contain two skill-related dimensions for workers, one dimension that encourages the development of related technology skills and a second dimension that addresses adjustment problems for workers that have been displaced by technological change.

The majority of studies evaluating ICT impacts on regional economies deal largely with computer hardware (Dedrick, Gurbaxani, and Kraemer 2003). Given

this emphasis, the impacts of other aspects of recent technological change, like broadband, are comparatively understudied. Although broadband has received little attention in ICT studies, there are some studies that have considered human capital impacts of broadband with an emphasis on employment impacts (Gillett et al. 2006; Crandall, Lehr, Litan 2009; Shideler, Badasyan, and Taylor 2007; Thompson and Garbacz 2008). These studies largely find a positive impact of broadband on employment levels.

In reality, however, the employment impacts of broadband on regional economies are a matter of some theoretical debate. This debate centers on skill-related impacts of technological change. For example, Jorgenson (2001) suggests the more recent divergence in the wages of less-educated and more educated workers is related to technological change. This divergence occurs because IT is a complimentary factor for highly skilled workers and a substitutable factor for low-skilled workers. In a related study, Autor, Levy, and Murnane (2003) found that the impacts of computers on workers were related to their type of work and type of worker. Combined, these studies suggest there are heterogeneities associated with broadband impacts on employment. These skill-related heterogeneities are also likely to exist for broadband-related impacts on productivity.

Data

This study evaluates spatial variations in the impact of broadband provision in 1999 on productivity changes between 2000 and 2007 for 3,046 counties in the continental United States (i.e., excluding Alaska and Hawaii). It also evaluates whether the quality of human capital within counties has a magnifying effect on broadband-related productivity impacts. The study period was selected for several reasons. First, it corresponds to the time period for which broadband data became available from the Federal Communications Commission (FCC). Second, it is immediately after the boom in telecommunications investments of the 1990s (Lowenstein 2004). The first consideration is particularly important in the context of this study because the initial distribution of broadband provision, which will be highlighted in this section, may have endowed regions with the necessary human capital resources to exploit this technology, with a first-mover productivity advantage in later years. The results of this study certainly suggest this is the case.

In the regression models specified in the next section, the dependent variable is the change in the natural logarithm of earnings in 2000 and the natural logarithm of earnings in 2007.¹ Earnings data were obtained from the BEA and consist of net earnings by place of residence² and income earned from interest, dividends, and rent (Regional Economic Information System [REIS] 2007). Earnings are used as a proxy for county-specific productivity measures because productivity data are not available at the county level. The most disaggregate level for which productivity data are available is the metropolitan area data reported by the Bureau of Economic Analysis (BEA). Given this data availability issue, an income-based measure of

productivity is used because GDP is comprised largely of earnings (64 percent of GDP), and therefore provides a reasonable estimate of output for most regions (BEA 2009).

Figure 1a and 1b shows the distribution of our dependent variable. Figure 1a highlights counties with a positive change in the dependent variable and Figure 1b highlights counties with a negative trend in earnings over the study period. Counties with a positive change in earnings are located primarily in the Western and Southern portions of the country. The Megalopolis portion of the country, particularly the Washington, DC, area, also displays a positive change in earnings. Counties with a negative change in earnings also exhibit a distinct spatial pattern and are concentrated around the Rust-belt, particularly in states such as Illinois, Indiana, Michigan, and Ohio.

In the econometric models that will be described in the next section, the two primary variables of interest are the sign and significance on the coefficient for broadband and the interaction variable of broadband with different measures of human capital. Broadband data were obtained from the Federal Communications Commission (FCC) Form 477 database. This national database compiles information from facilities-based providers that provide broadband on a variety of platforms including cable, xDSL, and wireless to report the number of broadband service providers in each ZIP code (FCC 2012). The FCC defines broadband as an always-on Internet connection that permits users to send and/or receive data using the Internet at transmission rates of *greater than* 200 kilobits per second in at least one direction (FCC 2012)³. Although more current data are available from the National Telecommunications and Information Administration from the National Broadband Mapping project, the historical 1999–2004 database was used as we were interested in evaluating the impact of the *initial* availability of broadband on productivity. Aside from this research goal, any study evaluating broadband impacts is necessarily a snapshot at one point in time because the upload and download speeds that define broadband are a moving target as technology improves over time (Grubestic 2012).

Two specifications of broadband are considered, a binary measure and a count measure. The binary measure considers the impact of the *presence* of broadband on productivity irrespective of “intensity”. A positive and significant coefficient on the binary measure of broadband would suggest that it is not necessarily the quantity of broadband providers that impacts productivity but rather the *potential* availability of high-speed Internet connections within a county. The count measure of broadband is the sum of the total number of broadband providers in that county, which is a measure of broadband availability used in several broadband-related studies (Grubestic and Murray 2004; Gillett et al. 2006; Grubestic 2006; Grubestic 2008; Mack and Grubestic 2009; Mack, Anselin, and Grubestic 2011; Mack 2012; Kolko 2012). While this measure does not necessarily capture the number of *unique* providers in a county, it is important because it indirectly captures some of the market dynamics associated with multiple providers in an area. More competitive broadband markets are associated with higher broadband speeds and better service quality (Kolko 2012). Higher broadband speeds offered in competitive markets are crucial

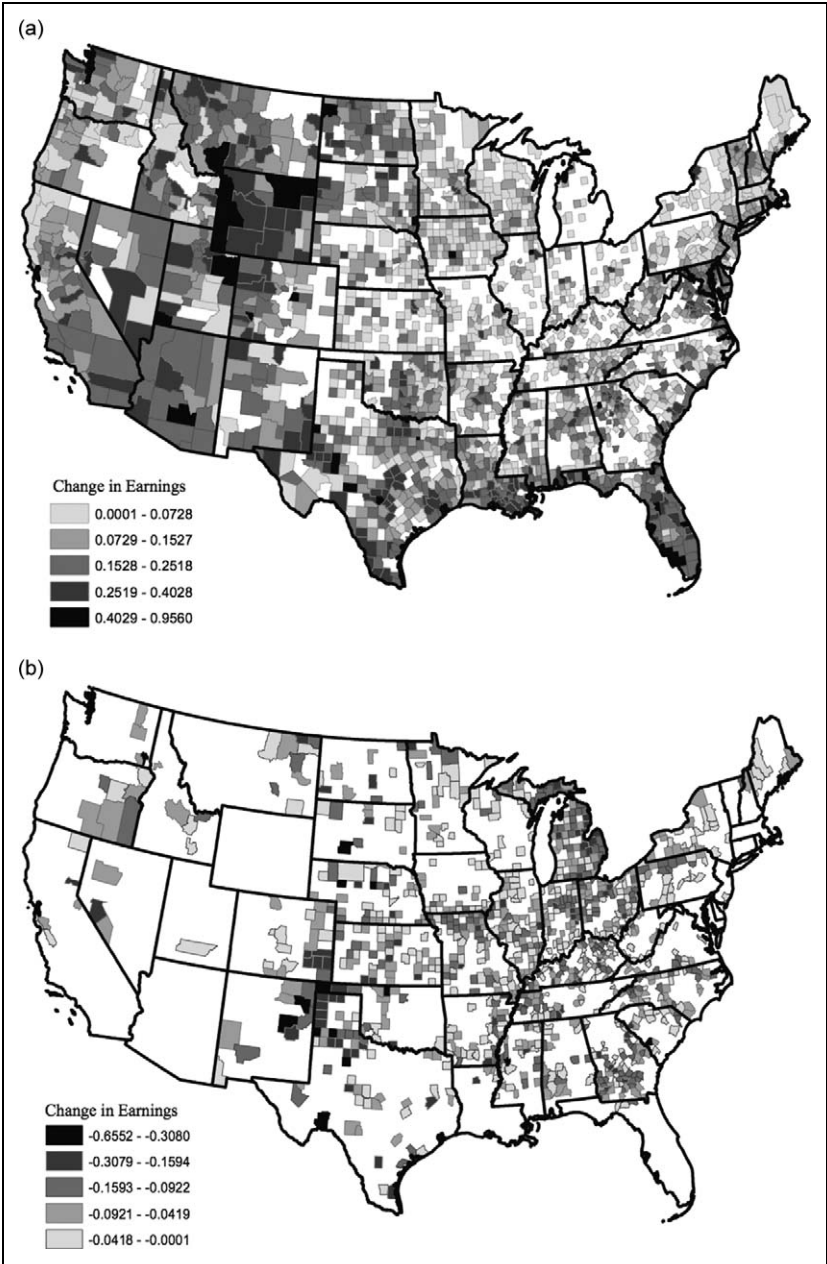


Figure 1. (a) Counties with a positive change in earnings (2000–2007) and (b) counties with a negative change in earnings (2000–2007).

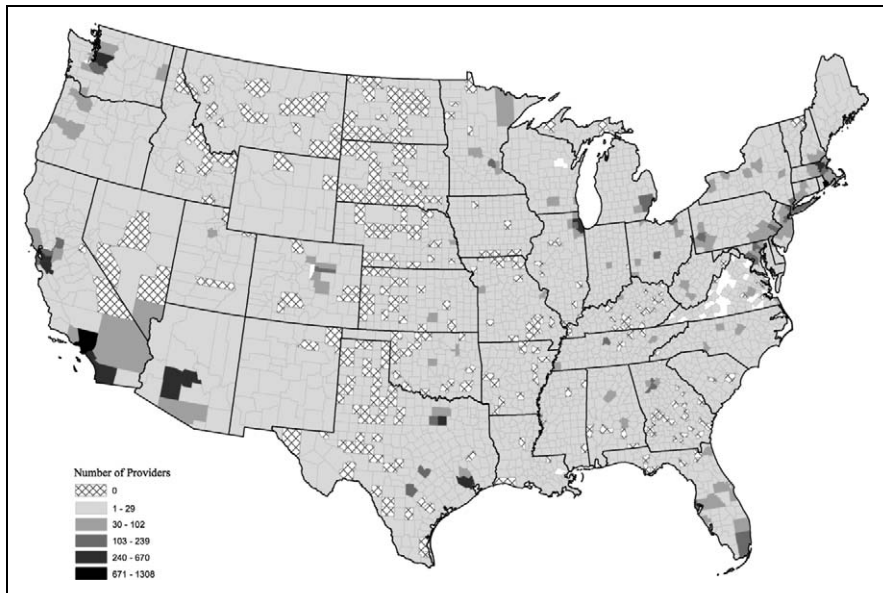


Figure 2. Distribution of broadband provision in 1999.

in the context of this study because they offer a wider variety of applications that may be linked to increased productivity during the study period of interest.

Figure 2 highlights the spatial distribution of broadband provision in 1999. It illustrates that most counties had at least one provider in this year with the exception of 351 counties. Counties with no broadband are located in the middle portion of the country in states such as Montana, North Dakota, South Dakota, and Kansas. Kentucky and Georgia also contain several counties with no broadband. This figure also highlights places with a large number of broadband providers. These counties are found largely in and around major metropolitan areas such as San Francisco, Chicago, Seattle, New York, and Boston.

Aside from broadband, other key variables in the model that are a focus of this study are those dealing with the quality of the labor force across counties. Labor force quality is specified in the models in two ways, an education variable and an occupational variable. The educational attainment of the labor force is specified as the percentage of people with a bachelor's degree or higher (US Census Bureau 2000). An occupational variable that measures the presence of workers in highly skilled occupations is also included in the model, in addition to the variable for educational attainment, because educational attainment does not necessarily capture the skill or the experience of the labor force (Florida, Mellander, and Stolarick 2008). The occupational variable used in this study is the number of people in highly skilled occupations specifically, management, professional, and related occupations (National

Historical Geographic Information System [NHGIS] 2012). It includes people working in the following occupations: computers, math, architecture, engineering, life, physical, and social sciences, community and social services, legal, education, training, and library, arts, design, entertainment, sports and media, health care, and farm management.

In addition to these measures of labor force quality, interaction variables are included in the models to measure not only the impact of broadband on productivity but also the link between broadband and the availability of potential users of broadband. As mentioned previously, prior studies of technological change found variability in computer impacts by type of work and type of worker (Autor, Levy, and Murnane 2003). They have also found higher use of computer technology by skilled workers (Entorf and Kramarz 1998). Similar variability is expected with respect to broadband-related impacts on productivity across different kinds of workers. The first interaction term multiplies the dummy variable for broadband presence with educational attainment. It is computed by multiplying the dummy variable for broadband by the percentage of people with a bachelor's degree or higher. The second interaction term considered is computed by multiplying the dummy variable for broadband with the variable for people working in highly skilled occupations. Although these interaction variables cannot capture the actual use of broadband by individuals, it can capture the *potential* use of broadband by people in each county. A positive and significant coefficient on these interaction terms indicates that the impact of broadband on productivity is magnified by the educational attainment of the workforce and/or the presence of people working in highly skilled occupations.

In addition to the primary independent variables of interest, several other independent variables that capture a variety of county-specific characteristics are used as control variables in the models. These variables control for a variety of characteristics that range from the level of infrastructure, the diversity of employment, race makeup, and the poverty level of counties. State-fixed effects are also included in the models to control for any state-specific characteristics that might impact productivity in the period of interest. These state-specific characteristics range from tax rates, to unionization issues, as well as weather. Table 1 contains a list of variables included in the models with their definitions and sources. Appendix A contains the descriptive statistics for these variables and Appendix B contains the correlation matrix for these variables.

Models

Two modeling approaches are common in studies evaluating the link between ICTs and productivity: production function models and growth accounting (Dedrick, Gurbaxani, and Kraemer 2003). Production function models are more common in firm-level studies while the growth accounting approach is used in economy-wide and industry-level studies (Dedrick, Gurbaxani, and Kraemer 2003). Although production

Table 1. Description of Variables.

Variable	Type of variable	Definition	Source
Change in earnings by place of residence (2000–2007)	Dependent	The change in earnings by place of residence between 2000 and 2007. Earnings by place of residence is defined as the natural logarithm of net earnings by place of residence plus dividends, interest, and rent. All earnings data are deflated to 2002 dollars	Regional Economic Information System (REIS), Bureau of Economic Analysis
Small- and medium-sized businesses	Independent	Establishments with fewer than 100 employees in 2000	County Business Patterns, US Census Bureau
Persons ages 18–64	Independent	Number of people between the ages of 18 and 64 in 2000	National Historic Geographic Information System (NHGIS)
Percent bachelor's or higher	Independent	Percentage of the population with a bachelor's degree or higher in 2000	US Census Bureau
Highly skilled occupations	Independent	The natural logarithm (ln) of people employed in management, professional, and related occupations in 2000	NHGIS
Herfindahl index	Independent	Herfindahl index created from the Standard Industrial Classification System employment in 2000	REIS
Government share of employment	Independent	Share of government employment across all SIC industries in 2000	REIS
Population density	Independent	Population density in 2000	US Census Bureau
Highway miles	Independent	Ln of total number of highway miles in a county in 2000. According to National Transportation Atlas Database (NTAD), highways are comprised of the following: rural arterials, urban principal arterials, and all National Highway System routes	NTAD, Bureau of Transportation Statistics
Broadband variable	Independent	Number of broadband providers in a county in 1999	Federal Communications Commission (FCC)

(continued)

Table 1. (continued)

Variable	Type of variable	Definition	Source
Broadband dummy variable	Independent	Whether a county had at least one broadband provider in 1999	FCC
Unemployment rate	Independent	Percentage of the population that is unemployed in 2000	Local Area Unemployment Statistics (LAUS) Program from the Bureau of Labor Statistics
Percent black	Independent	Percentage of the population that is black in 2000	NHGIS
Total poverty rate	Independent	Percentage of people living in poverty in 2000 as defined by the US Census Bureau	US Census Bureau
Presence of a research university	Independent	Dummy variable that indicates whether a county contains a research I university, as defined by the Carnegie Classification System in 2000	Integrated Postsecondary Education Data System
Metropolitan area membership in 2000	Independent	Dummy variable that indicates whether a county was a member of a metropolitan area in 2000	US Census Bureau
Urban area membership in 2000	Independent	Dummy variable that indicates whether a county contains an urbanized area in 2000	US Census Bureau
Broadband dummy interaction with bachelor's degree or higher	Independent	Interaction variable computed by multiplying the dummy variable for broadband in 1999 with the percentage of people with a bachelor's degree or higher in 2000	FCC and the US Census Bureau
Broadband dummy interaction with highly skilled occupations	Independent	Interaction variable computed by multiplying the dummy variable for broadband in 1999 with the ln of people employed in highly skilled occupations in the year 2000	FCC and the US Census Bureau

function and growth accounting models are the more popular approaches to modeling ICT-related impacts on productivity, a multitude of modeling approaches have been utilized to evaluate the link between ICTs and productivity. These approaches include dynamic general equilibrium models (Martínez, Rodríguez, and Torres 2010), panel data approaches (Majumdar 2008; Majumdar, Carare, and Chang 2010), simultaneous equation modeling (Koutroumpis 2009), and ordinary least squares regression (Brynjolfsson and Hitt 1996; Basu et al. 2004). While these studies consider a variety of issues associated with modeling this relationship, including simultaneity issues (Kolko 2012), a notable gap in these studies is their explicit attention to spatial effects.

The modeling approach proposed in this study is designed to address this gap and explicitly address county-level variations in productivity that are related to the variables described above, and also the well-noted heterogeneous distribution of broadband provision across space (Grubestic and Murray 2002; Grubestic 2003; Mack and Grubestic 2009). Given the host of issues associated with unaddressed spatial effects, including biased coefficient estimates and standard errors, an explicit consideration of space in the modeling process is critical. The model development process considered two kinds of spatial models to account for spatial structure in the data, a spatial lag model and a spatial error model. These are specified in equations (1) and (2) below (Anselin 1988):

$$y = \rho Wy + X\beta + \varepsilon_1, \quad (1)$$

$$y = X\beta + \lambda W\varepsilon_2 + \mu, \quad (2)$$

where:

ρ = the amount of autocorrelation in the dependent variable.

X = a matrix of independent variables.

W = a spatial weights matrix that quantifies the spatial structure to the data.

β = a matrix of coefficients for each of the independent variables.

ε_1 = the error term in the spatial lag model which is identically distributed (i.i.d).

λ = the amount of autocorrelation present in the error term.

ε_2 = the autocorrelated portion of the error term in the spatial error model.

μ = the error term in the spatial error model which is i.i.d.

The spatial lag specification models explicit autocorrelation in the dependent variable while the spatial error model deals with autocorrelation of a nonspecific nature. The Anselin–Kelejian test (Anselin and Kelejian 1997) will be used to determine whether a lag or error specification is best suited for this study. Other modeling issues addressed by the model development process include biased coefficients due to potential simultaneity between broadband and productivity and heteroscedastic errors. The Wu–Hausman test⁴ (Hausman 1978; Wu 1973) is used to diagnose any simultaneity between the count version of the broadband variable and productivity. Two corrections for heteroscedasticity are also considered, the White correction (White 1980) and the heteroscedasticity-autocorrelation consistent (HAC) correction (Kelejian and Prucha 2007). The White correction adjusts the standard errors

for unknown causes of heteroscedasticity while the HAC correction is able to account for both unknown causes of heteroscedasticity and any remaining autocorrelation not accounted for by including a spatial lag term in the model. The results of the Wu–Hausman test revealed no simultaneity between broadband and productivity and the test statistics illustrating this absence of simultaneity, where relevant, are included in the tables presenting the modeling results in the next section. The Anselin–Kelejian test also suggested a lag spatial specification was most appropriate to model the relationship of interest and that the lag term, in addition to the other spatial variables in the model fully accounted for the autocorrelation present in these data. Given these two sets of test results, the models⁵ discussed in the next section are produced via a spatial lag specification with White corrected standard errors.⁶

Results

As described in the Data section, several model specifications were considered to evaluate the impact of broadband on productivity, and whether or not human capital plays a role in this impact. The model results are displayed in Tables 2 and 3. The specifications vary across three dimensions: the measure of broadband included in the model, the measure of labor force quality included in the model (educational or occupational), and the interaction terms between broadband and human capital included in the model. Both sets of models incorporate different measures of educational attainment and occupational skill as both independent variables and interaction variables; the key difference between the models in Tables 2 and 3 is the specification of broadband. Table 2 contains different model specifications where broadband is included as a dummy variable. Table 3 contains a variety of model specifications where broadband is incorporated as a count variable. The other key difference between Tables 2 and 3 is the exclusion of small- to medium-sized businesses from Table 3 because of multicollinearity issues with the count version of the broadband variable.⁷

Control Covariates

Irrespective of the specification of broadband and labor force quality, the other covariates in Tables 2 and 3 are fairly consistent in their sign, size, and significance. In Table 2, small- to medium- sized businesses have a negative impact on productivity. This result is consistent with prior studies, which found a positive correlation between firm size and wages, or that smaller firms pay lower wages than large firms (Lester 1967; Masters 1969; Antos 1981; Mellow 1982; Atrostic 1983; Oi 1983; Evans and Leighton 1989; Reilly 1995). This employer-size wage effect states that establishment size is a proxy for unobserved levels of human capital across establishments of various sizes (Hamermesh 1980). Large firms are able to pay workers higher wages because they are able to attract higher quality workers, have the financial means to pay workers more, are less able to monitor their workers, and offer

Table 2. Model Results with Broadband Included as a Dummy Variable.

	Model 1	Model 2	Model 3	Model 4
Constant	-0.3200***	-0.3882***	-0.2450***	-0.2638***
Small- and medium-sized businesses	-0.0000009***	-0.0000014***	-0.0000009***	-0.0000013***
Persons ages 18-64	0.3008***	0.2643***	0.2746***	0.2711***
Metropolitan area membership in 2000	0.0338***	0.0156***	0.0332***	0.0200***
Percent bachelor's or higher	0.0059***	0.0047***	-	0.0049***
Presence of a research university	-0.0321***	-0.0434***	-0.0329***	-0.04***
Urban area membership	0.0187***	0.0053	0.0178***	0.0103*
Herfindahl index	0.0670	0.0098	0.0667	0.0304
Government share of employment	0.0800**	0.1220***	0.0800**	0.1085***
Highway miles	0.0005	-0.0145***	0.0007	-0.0097**
Population density	0.0000031***	0.0000018	0.0000032***	0.0000022
Unemployment rate	-0.0007	-0.0003	-0.0013	-0.0004
Highly skilled occupations	-	0.0228***	-	-
Broadband variable	-	-	-	-
Broadband dummy variable	0.0230***	0.0140*	-0.0347***	-0.0915***
Broadband dummy interaction with bachelor's degree or higher	-	-	0.0062***	-
Broadband dummy interaction with highly skilled occupations	-	-	-	0.0166***
Percent black	-0.1115***	-0.1159***	-0.1108***	-0.1145***
Total poverty rate	-0.0004	-0.0003	-0.0004	-0.0003
Spatial lag	0.5445***	0.4786***	0.5442***	0.5112***
Wu-Hausman test statistic	N/A	N/A	N/A	N/A
Pseudo R ²	0.4701	0.4748	0.4715	0.4726
Spatial pseudo R ²	0.3945	0.4053	0.3961	0.4014
Degrees of freedom	2,983	2,982	2,983	2,982

*Significant at the 10 percent level.

**Significant at the 5 percent level.

***Significant at the 1 percent level.

Table 3. Model Results with Broadband Included as a Count Variable.

	Model 1	Model 2	Model 3	Model 4
Constant	-0.3238***	-0.3923***	-0.2791***	-0.3060***
Small- and medium-sized businesses	—	—	—	—
Persons ages 18–64	0.3337***	0.2876***	0.3181***	0.2915***
Metropolitan area membership in 2000	0.0337***	0.0143**	0.0362***	0.0300***
Percent bachelor's or higher	0.0059***	0.0045***	—	0.0057***
Presence of a research university	-0.0337***	-0.0464***	-0.0291***	-0.0351***
Urban area membership	0.0230***	0.0072	0.0133***	0.0151***
Herfindahl index	0.0761	0.0118	0.0754	0.0542
Government share of employment	0.0730**	0.1221***	0.0775**	0.0915***
Highway miles	0.0013	-0.0154***	-0.0007	-0.0029
Population density	0.0000025***	0.0000005	0.0000025***	0.0000020**
Unemployment rate	-0.0004	0.0002	-0.0028	-0.0007
Highly skilled occupations	—	0.0236***	—	—
Broadband variable	-0.0002**	-0.0002**	-0.0002**	-0.0002**
Broadband dummy variable	—	—	—	—
Broadband dummy interaction with bachelor's degree or higher	—	—	0.0047***	—
Broadband dummy interaction with highly skilled occupations	—	—	—	0.0043***
Percent black	-0.1100***	-0.1102***	-0.1102***	-0.1130***
Total poverty rate	-0.0006	-0.0004	-0.0003	-0.0003
Spatial lag	0.5318***	0.4672***	0.5365***	0.5254***
Wu-Hausman test statistic	0.1170	0.0486	0.0222	0.0735
Pseudo R ²	0.4675	0.473	0.4686	0.471
Spatial pseudo R ²	0.3921	0.4054	0.3951	0.3992
Degrees of freedom	2,984	2,983	2,984	2,983

*Significant at the 10 percent level.

**Significant at the 5 percent level.

***Significant at the 1 percent level.

inferior working conditions that must be compensated more with higher wages (Brown and Medoff 1989).

Other variables that have a negative and significant effect on productivity are the presence of a research university, percent black, and the number of highway miles. College towns are fairly well known for their lower per capita income compared to places without colleges and universities. Therefore, the negative impact of research universities reflects this trend and speaks to the large student population employed in low-wage, part-time jobs that receive financial assistance from their parents (Gumprecht 2008). The negative coefficient on highway miles likely reflects the large number of highways in the west and more rural areas of the country where labor markets are smaller and less competitive than in large urban centers.

Several variables had a positive and significant impact on productivity. These include labor force size (persons 18–64), metropolitan area membership, urban area membership, population density, educational attainment, the share of government employment, and the variable for highly skilled occupations. These results are consistent with previous findings. The sign on labor force size, urban area membership, and metropolitan area membership, for instance, are consistent with the findings of Rauch (1993) and Glaeser, Scheinkman, and Shleifer (1995) who found that wages and income per capita, respectively, and grow faster in cities, which have higher levels of human capital spillovers. The positive sign on educational attainment and skilled occupations are also consistent with the general human capital theory and the expectation that people with higher levels of educational attainment and skills are more productive (Becker 1964).

Broadband

Unlike the control covariates in the model, the sign on the coefficient for broadband exhibits marked variation across the eight models in Tables 2 and 3. This variation is related to whether broadband is specified as a dummy or a count variable and also the presence of interaction variables between broadband and different measures of human capital. Across the eight models, however, the results all suggest that broadband alone is not responsible for increased levels of productivity, but rather the availability of both broadband and high-quality human capital is key to broadband-related productivity increases.

Table 2 contains a series of models where broadband is included as a dummy variable. When specified in this manner, broadband provision has a positive impact on productivity. This result suggests that the presence of broadband and not the quantity of broadband in regions is what impacts productivity in future years. The positive sign on the broadband dummy variable changes when interaction variables with educational attainment and highly skilled occupations are included in the model. When these interaction terms are included, the sign on broadband becomes negative and the interaction terms of broadband with some measure of higher quality human capital (educational attainment in model 3 and highly skilled occupations in model 4) is positive and significant.

This change in sign when controlling for the joint presence of broadband and human capital provides support for the findings of prior studies, which find that highly skilled workers are more likely to use computer technology (Entorf and Kramarz 1998). They also suggest that people with at least a bachelor's degree are more capable of exploiting broadband for productive uses, perhaps because of exposure to different Internet applications on college campuses and/or training in the classroom. Together, models 3 and 4 in Table 2 suggest that while the presence of people with higher skill levels and educational attainment are important to productivity alone, broadband only has an impact on productivity when regions also contain high levels of human capital.

The negative sign on broadband in these two models as well as across all the models in Table 3 provide support for studies that suggest broadband does not necessarily translate into productive uses (Gordon 2000; Kolko 2010). Kolko (2010) found, for example, that a switch from dial-up to broadband Internet connection speeds only increases the use of particular applications such as downloading music and online shopping, which are not necessarily productive uses of the Internet. This is somewhat of a surprise, given the association of higher levels of broadband with competitive markets (Kolko 2012), and the associated benefits of better service quality and higher broadband speeds that benefit consumers in competitive markets. In this regard, consumers are anticipated to benefit from not only lower prices for broadband service but also a wider array of broadband applications that are possible in competitive markets with higher broadband speeds (Ezell et al. 2009). For example, productivity gains may be expected from real-time videoconferencing in place of commuting for face-to-face communications, as well as faster transfer of large volumes of data.

The negative and significant sign on broadband in combination with the positive and significant signs on the broadband interaction variables with educational attainment and highly skilled occupations suggest instead that the faster Internet connection speeds made possible by broadband reduce productivity unless used by people with a priori training in productive uses of this application and/or with the requisite business experience to understand how broadband may be effectively incorporated in their work processes. This evidence of skill-based technological change also supports the findings of Autor, Levy, and Murnane (2003) who found that the impacts of computers on workers were related to the type of work performed and type of worker. They find that computer capital functions as a substitute for routine and manual tasks and functions as a complement for nonroutine and complex tasks.

Spatial Effects

A final model result worthy of mention is the presence of rather large and significant spatial effects in Tables 2 and 3. The presence of these spatial effects not only highlights the importance of the explicit spatial modeling approach used in this article, but it also identifies spillover effects in the independent variables above and beyond the marginal coefficient estimates presented in these two tables. To determine the

total effect of each independent variable in the model, inclusive of the positive spillovers indicated by the lag term, the marginal coefficient estimate for each independent variable may be multiplied by $1/(1 - \rho)$ (Mobley et al. 2006). If this is done for the coefficient estimate on broadband, the total effect of broadband on productivity in Table 2 ranges from 0.0269 to 0.0505 with no human capital interaction variables in the model and -0.1872 to -0.0761 with human capital interaction variables included in the model. In Table 3, the coefficient estimate on broadband was robust across all models, and the total effect of broadband in these models is about -0.0004 . This approach yields a higher impact of broadband on county productivity and highlights the magnifying impact of spatial spillovers.

Discussion and Conclusion

The goal of this study was to evaluate the link between broadband and productivity for the 2000–2007 period. This productivity link was explored across two dimensions, the impact of broadband alone, and the combined impact of broadband and labor force quality. The models developed in this study diverged from prior assessments of ICT-related productivity impacts by their explicit consideration and incorporation of spatial effects in the modeling process. This modeling approach revealed several interesting aspects about the impact of broadband and productivity, and, also suggested several areas for future research.

Perhaps the most interesting aspect of this study was the finding of a skills specific link to broadband-related productivity impacts. When controlling for educational attainment, highly skilled occupations, and the other covariates mentioned previously, broadband alone does have a positive impact on productivity. When either of the two interaction variables that evaluate the impact of broadband provision presence and people with a bachelor's degree or higher or broadband provision presence and the number of highly skilled people jointly, this positive sign becomes negative. Although the interaction variable only captures *potential use* of broadband, this change in sign on broadband suggests that broadband only produces positive productivity impacts when used by more educated and/or highly skilled occupations. The addition of an interaction term separates out educated people and skill-related technology benefits from the coefficient on broadband, which appears to account for all the positive benefits broadband provision has on productivity. This separation of effects is why the coefficient for broadband becomes negative when interaction variables are added to the models.

This separation of effects means the level of broadband provision in counties in 1999 had a negative impact on productivity for the 2000–2007. Given the multitude of Internet applications that can be exploited with high-speed broadband connections, this result seems counterintuitive at first glance. However, there are several plausible explanations for this negative impact. One explanation is a divergent impact on worker wages related to the skill level of workers. Jorgenson (2001) has suggested technological change as an explanation of the visible divergence in the

wages of highly skilled and low-skilled workers. This divergence occurs because IT is a complement for highly skilled workers and a substitute for low-skilled workers (Autor, Levy, and Murnane 2003).

Other explanations for this negative sign on broadband are also possible. The negative coefficient could indicate a substitution of technology for workers, which corresponds with the hypothesis of prior studies that have suggested cheaper technology could lead to the use of more technology in place of expensive workers (Kolko, 2012). More broadband providers is an indicator of more competitive broadband markets, and thus, the lower costs and wider range of applications possible in competitive areas could lead businesses to substitute technology for labor in this time period. A related effect could be a lower wage for workers that were put at a relative skill disadvantage with the arrival of this technology. These workers may have seen a decline in their income over the 2000–2007 period after the arrival of broadband because of a relative lack of skill. This negative impact on less educated and lower skilled workers may be exacerbated because of the relative immobility of lower skilled workers compared to higher skilled workers, which is consistent with the human capital migration theory (Sjaastad 1962; Faggian and McCann 2006; Faggian, McCann, and Sheppard 2007).

From a policy standpoint, this evidence of skill-based technological change suggests ICT policies should incorporate a training aspect to encourage the development of technology skills. To date, national broadband policy has focused on improving competition between private sector providers (Telecommunications Act of 1996). The model results suggest, however, that developing policies and programs that make *people* more competitive as well will improve regional productivity and prosperity. In fact, the more recently devised National Broadband Plan recognizes that utilization, or the intensity and quality of use of Internet connections for personal and business purposes, impacts the offline benefits of users (FCC 2012). The results of this study suggest that people with better broadband utilization have the largest positive impacts on county level productivity. In this regard, broadband policies should explicitly address effective utilization of high-speed Internet connections. Parham (2002) in particular suggests ICT policies should contain two dimensions, one that promotes technology skills and a second dimension that addresses adjustment problems for workers that have been displaced by technological change. Thus, policies that address broadband access to the exclusion of adoption and utilization issues are missing a key piece of the puzzle.

While the results of the regression analysis suggest there is a skill-related link between broadband and productivity, more work is required to examine the generalizability of these findings to other time periods. This study examined the initial impacts of broadband in 1999 on productivity in counties over the 2000–2007 study period and found a generally negative impact. In addition to the previously mentioned disparate impacts on workers of different skill levels, the negative sign could reflect start-up costs associated with learning to use this new technology. Thus, it would be interesting to evaluate how this link changes or does not change

when examined in later time periods. Although not discussed here, the change in 2000–2004 earnings was also considered as dependent variable to test the immediacy of the impact of broadband availability on productivity. Broadband in 1999 in these models was insignificant and thus had no impact on productivity in this time period.

This result for the 2000–2004 study period suggests time is a key element for productivity impacts to be realized from new technologies. In 1999, broadband and Internet applications of any kind were a rather novel concept; the capabilities of which were not widely understood by the general public. The positive and significant interaction on the education and skills interaction variables suggest that places with an initial endowment of broadband and people that knew how to effectively exploit this technology had a first mover advantage in productivity improvements over the next several years. It would be interesting to evaluate this relationship over the next several years, as the data become available, to determine how the diffusion of broadband and the familiarity with its possibilities impacted productivity.

If the negative sign is indicative of start-up costs and costs associated with learning related to new technologies, then it would be expected that the negative sign on broadband might switch to a positive sign as familiarity with high-speed Internet applications increases in later years. If the negative sign is entirely related to disparate impacts by worker skill levels and education, then the results of this analysis would be expected to persist for evaluations of this relationship in subsequent time periods. Another extension to this study that would add further insight about broadband-related productivity improvement is a breakdown of productivity impacts by people working in different industries to see if broadband increases productivity in some industries more so than others. Several studies have noted an uneven adoption of dial-up Internet connections across industries (Forman, Goldfarb, and Greenstein 2005), and this uneven adoption may likely impact productivity levels in these industries.

The results of this study certainly represent an important step in unraveling the impact of broadband Internet connections on regional economies. Access to and utilization of technology is an increasingly key component to the success of businesses and individuals nationwide. However, much work remains to unravel the impacts of new technologies on the competitiveness of people and businesses, and understand what these impacts mean for regional development trends. As the rate of technological change increases, this historical snapshot of broadband-related productivity impacts provides important information about how future technological innovations may impact regional productivity.

Appendix A

Descriptive Statistics

Variable	Observations	Mean	SD	Min	Max
Change in earnings by place of residence (2000–2007)	3,046	0.06	0.13	–0.66	0.96
Small- and medium-sized businesses	3,046	2,206.57	7,393.84	1.00	220,480.00
Number of broadband providers in 1999	3,046	9.54	38.22	0.00	1,308.00
Persons ages 18–64	3,046	0.60	0.04	0.46	0.79
Herfindahl index	3,046	0.15	0.05	0.01	0.81
Government share of employment	3,046	0.17	0.07	0.04	0.90
Ln highway miles	3,046	4.72	0.64	0.00	7.92
Unemployment rate	3,046	4.36	1.64	1.40	17.40
Highly skilled occupations	3,046	8.13	1.49	2.40	14.12
Poverty rate	3,046	13.95	5.24	0.00	46.00
Population density	3,046	178.81	1,160.41	0.10	45,519.54
Percent black	3,046	0.09	0.14	0.00	0.86
Percent bachelor's or higher	3,046	10.89	4.87	0.00	40.02
Broadband dummy interaction with bachelor's degree or higher	3,046	9.80	5.85	0.00	40.02
Broadband dummy interaction with highly skilled occupations	3,046	7.39	2.98	0.00	14.12

Appendix B Correlation Matrix

	Change in earnings by place of residence (2000-2007)	Small- and medium-sized businesses	Persons ages 18-64	Percent bachelor's or higher	Highly skilled occupations	Herfindahl index	Government share of employment	Population density	Highway miles	Broadband variable	Unemployment rate	Percent black	Total poverty rate	Broadband dummy interaction with bachelor's degree or higher	Broadband dummy interaction with highly skilled occupations
Change in earnings by place of residence (2000-2007)	1														
Small- and medium-sized businesses	0.1089	1													
Persons ages 18-64	0.1796	0.188	1												
Percent bachelor's or higher	0.3683	0.333	0.3077	1											
Highly skilled occupations	0.2722	0.5614	0.4628	0.5107	1										
Herfindahl index	0.1378	0.1909	0.4236	0.2263	0.4261	1									
Government share of employment	0.0389	-0.1345	0.0533	-0.1148	-0.2341	0.2065	1								
Population density	0.047	0.438	0.1364	0.1652	0.2897	0.1448	-0.0628	1							
Highway miles	0.2044	0.4084	0.0911	0.3002	0.587	0.2083	-0.0208	0.0967	1						
Broadband variable	0.0764	0.951	0.1611	0.2813	0.4716	0.1554	-0.1137	0.3064	0.3552	1					

(continued)

Appendix B. (continued)

	Change in earnings by place of residence (2000- 2007)	Small- and medium- sized businesses	Persons ages 18-64	Percent bachelor's or higher	Highly skilled occupations	Herfindahl index	Government share of employment	Population density	Highway miles	Broadband variable	Unemployment rate	Percent black	Total poverty rate	Broadband dummy interaction with bachelor's degree or higher	Broadband dummy interaction with highly skilled occupations
Unemployment rate	-0.1222	-0.0703	-0.0776	-0.4271	-0.1857	0.0495	0.2815	-0.0089	0.0181	-0.0615	I				
Percent black	-0.0343	0.0652	0.1584	-0.1297	0.0633	0.1598	0.1754	0.0953	-0.0051	0.0463	0.2435	I			
Total poverty rate	-0.0783	-0.1094	-0.0881	-0.3371	-0.2916	-0.029	0.3448	-0.0267	-0.076	-0.1049	0.5409	0.3532	I		
Broadband dummy interaction with bachelor's degree or higher dummy	0.3537	0.3294	0.4304	0.8243	0.651	0.3194	-0.1508	0.1648	0.3472	0.2809	-0.3193	-0.0512	-0.3342	I	
Broadband dummy interaction with highly skilled occupations	0.2141	0.3491	0.419	0.3314	0.7707	0.3623	-0.199	0.1799	0.4318	0.2974	-0.0962	0.0703	-0.255	0.7406	I

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Notes

1. This variable was deflated to 2002 dollars.
2. Both earnings by place of work and earnings by place of residence were considered as dependent variables. It was necessary to use earnings by place of residence, however, to avoid spatial mismatch between the dependent variable and the independent variables in the model. Most of the independent variables are place of residence based and, therefore, use of a place of work measure as the dependent variable is not appropriate for the purposes of this study.
3. ZIP code data from 1999 were aggregated to the county level by computing the centroid of the ZIP code and aggregating the centroids to county boundaries. ZIP code areas are not nested within counties and thus this aggregation routine was necessary to avoid any double counting of broadband providers due to the irregular shape of ZIP code areas.
4. The software Stata was used for the Wu–Hausman test.
5. Models were estimated in Geodaspace, which is an open source software developed by the Geoda Center for Geospatial Analysis and Computation. <https://geodacenter.asu.edu/geodaspace-mode>.
6. An eight nearest neighbor weights matrix was used to estimate these spatial models. This was done to ensure all counties had at least one neighbor in the modeling process. Other numbers of nearest neighbors (four and six) were considered to evaluate the robustness of models to different weights matrices. Model results were robust to different numbers of nearest neighbors.
7. The correlation between these two variables is 0.9510.

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